

Homework 2 Data Mining

Part 1: Reflections on Homework 1

From Homework 1, I learned the importance of storytelling through data, not just presenting charts and stats. While analyzing vehicle fuel consumption data, I gained valuable experience in profiling a large dataset, identifying key trends, and using interactive dashboards (Streamlit + Plotly) to communicate insights effectively.

One key takeaway was how visualizations can guide decision-making when supported by clean and well-explained data. I also learned that understanding the audience (non-technical stakeholders) is critical — insights should be presented clearly, with focused takeaways instead of overwhelming technical detail.

For future analysis, I plan to:

- Focus more on data cleaning and preparation, as it significantly improves downstream analysis
- Use feature importance metrics to guide storytelling in modeling tasks
- Emphasize interpretability when presenting models to stakeholders
- Continue developing interactive dashboards to give users control over insights

Homework 1 helped me grow in not just technical skills, but in the ability to turn raw data into meaningful narratives that drive business impact.

Part 2: Model Card

Property	Decision Tree	Naive Bayes	K-Nearest Neighbor	Logistic Regression	Support Vector Machine (SVM)
Parametric/Non-parametric	Non-parametric	Parametric	Non-parametric	Parametric	Can be both
Input	Both	Discrete preferred	Both	Both	Both
Output	Discrete	Discrete	Discrete	Discrete	Discrete
Handle Missing Value	No	No	No	No	No
Model Representation	Tree structure	Probabilistic (Bayes)	Memory-based	Linear equation	Hyperplanes
Model Parameters	Max depth, min samples	Prior, likelihood	k	Coefficients, intercept	C, kernel, gamma

Make More Complex	Increase depth	Add features	Increase k	Add polynomial terms	Use RBF kernel
Make Less Complex	Prune tree	Feature selection	Decrease k	Regularization	Use linear kernel
Interpretability/Transparency	High	Medium-High	Low	Medium-High	Low

Part 3: Wine-Tasting Machine

1. HTML File

Overview

Brought to you by YData

OverviewAlerts4Reproduction

Dataset statistics

Number of variables	4
Number of observations	571
Missing cells	0
Missing cells (%)	0.0%
Duplicate rows	13
Duplicate rows (%)	2.3%
Total size in memory	42.8 KiB
Average record size in memory	76.8 B

Variable types

Numeric	3
Categorical	1

Variables

Variables

Select Columns ▾

citric acid

Real number (ℝ)

Zeros

Distinct	74	Minimum	0
Distinct (%)	13.0%	Maximum	1
Missing	0	Zeros	25
Missing (%)	0.0%	Zeros (%)	4.4%
Infinite	0	Negative	0
Infinite (%)	0.0%	Negative (%)	0.0%
Mean	0.32467601	Memory size	4.6 KiB

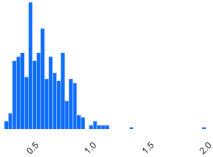
More details

sulphates

Real number (R)

High correlation

Distinct	77	Minimum	0.25
Distinct (%)	13.5%	Maximum	2
Missing	0	Zeros	0
Missing (%)	0.0%	Zeros (%)	0.0%
Infinite	0	Negative	0
Infinite (%)	0.0%	Negative (%)	0.0%
Mean	0.58816112	Memory size	4.6 KiB

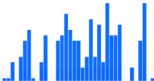


More details

alcohol

Real number (R)

Distinct	49	Minimum	8
Distinct (%)	8.6%	Maximum	12.8
Missing	0	Zeros	0



Interactions

citric acid

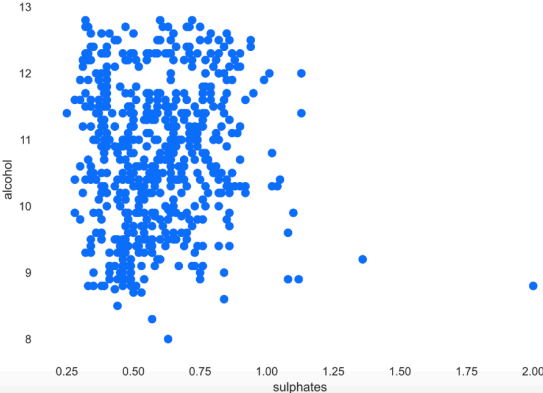
sulphates

alcohol

alcohol

citric acid

sulphates

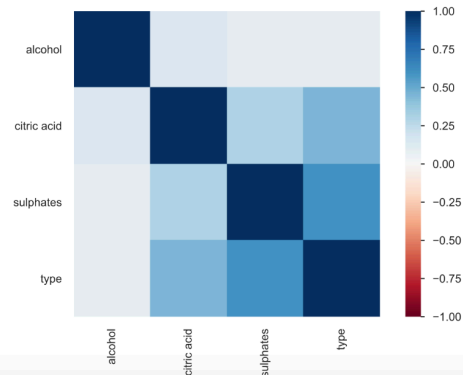


Correlations

Auto

Heatmap

Table



2.

<https://github.com/JanhaviJ006/jj/blob/main/HW2/HW2%20IS733.ipynb>

Baseline Accuracy: 52.89

Baseline AUC: 0.5

Model: Logistic Regression

Accuracy: 79.51

AUC: 0.8709470937246117

Model: Naive Bayes

Accuracy: 82.14

AUC: 0.882481104901647

Model: Decision Tree

Accuracy: 76.18

AUC: 0.7682550038159481

Model: SVM-Linear

Accuracy: 78.98

AUC: 0.8715010216893573

Model: SVM-RBF

Accuracy: 82.31

AUC: 0.9104975504074447

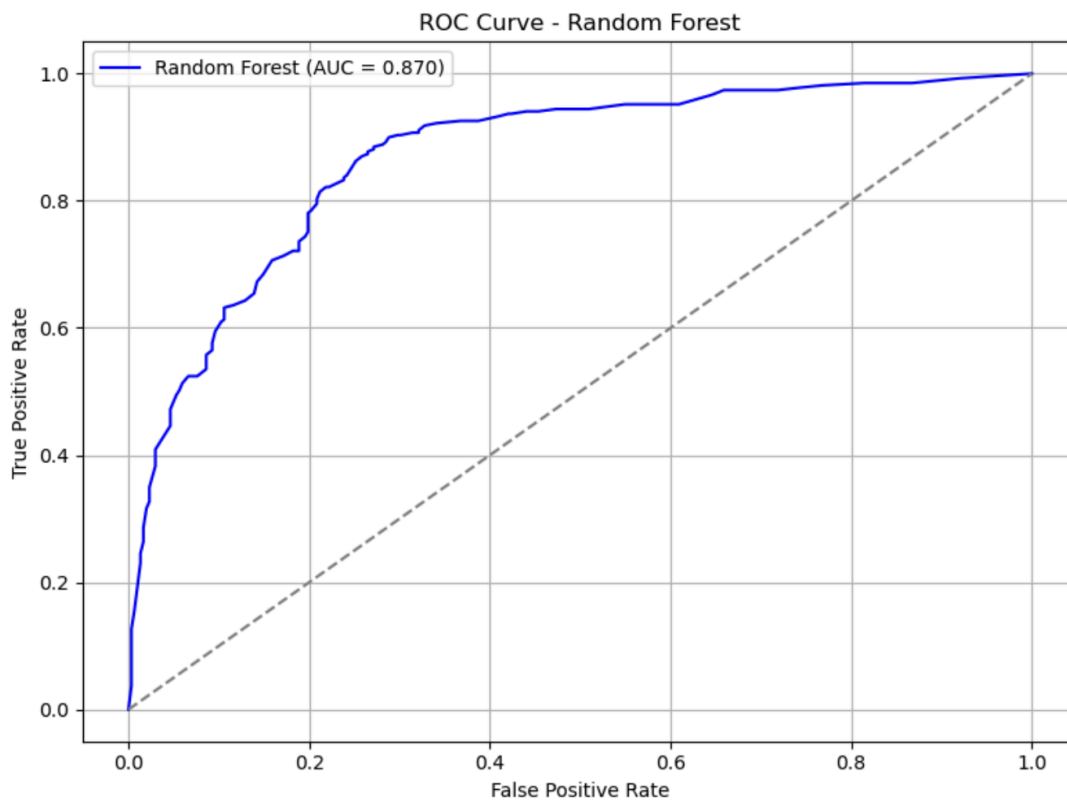
Model: Random Forest

Accuracy: 78.28

AUC: 0.866601836578941

Model	Baseline	Logistic Regression	Naive Bayes	Decision Tree	SVM-Li near	SVM-RB F	Random Forest
AUC	0.5	0.87095	0.88248	0.76826	0.87150	0.91050	0.86660
Accuracy	52.89%	79.51%	82.14%	76.18%	78.98%	82.31%	78.28%

3.



4. Accuracy: 81.25
AUC Score: 0.9455

The SVM-RBF model did a great job when tested on the white wine data. Even though it was trained only on red wine, it still got over 81% accuracy and a very high AUC of 0.95.

This means the model is really good at telling the difference between wine types based on just a few features like alcohol, sulphates, and citric acid.

The fact that it performed so well on new, unseen data shows that it's a strong and reliable model — and it could definitely be useful in real-world wine classification.

5. I would recommend using a Decision Tree or Logistic Regression model.

These models are much easier to understand and explain. For example:

- A Decision Tree shows a clear set of “if-then” rules (like “if alcohol > 10%, then likely red”), which is great for helping wine experts see how the model is making its decisions.
- Logistic Regression gives simple weights to each feature, so experts can see exactly which chemical properties (like sulphates or alcohol) are influencing the prediction the most.

Models like SVM or Random Forest might be more accurate in some cases, but they work more like black boxes — they make decisions without giving much insight into why.

So for wine-tasting experts who want to understand the reasoning behind predictions, simpler and more transparent models like Decision Trees and Logistic Regression are the better choice.